

Actividad 1 — Identificación de la fuente

Fuente: Our World in Data (OWID) — United Nations, World Population Prospects (2024)

Cita:

UN, World Population Prospects (2024) – processed by Our World in Data.
"Annual change in population – UN WPP" [dataset]. United Nations, "World Population Prospects"; United Nations, "World Population Prospects - Interim Update" [original data].

1. **¿Quién produjo originalmente los datos?** Las Naciones Unidas (UN), World Population Prospects (2024).
2. **¿Qué organización los procesa para su utilización en Our World in Data?** Our World in Data (OWID).
3. **¿Qué representa la variable que analizarás?** El cambio neto anual de población (nacimientos, defunciones y migración combinados).

Actividad 2 — Carga y exploración de datos

```
In [1]: import pandas as pd
import numpy as np

In [2]: df = pd.read_csv("../data/raw/annual-population-growth/annual-population-growth.csv")

In [3]: df.head()
```

	Entity	Code	Year	Annual change in population	Annual population change (Projected)
0	Afghanistan	AFG	1951	103163.0	NaN
1	Afghanistan	AFG	1952	108441.0	NaN
2	Afghanistan	AFG	1953	108919.0	NaN
3	Afghanistan	AFG	1954	111251.0	NaN
4	Afghanistan	AFG	1955	119027.0	NaN

```
In [4]: df.shape
(38400, 5)

In [5]: df.columns.tolist()
```

```
['Entity', 'Code', 'Year', 'Annual change in population', 'Annual population change (Projected)']
```

```
In [6]: df.dtypes
```

```
Entity                object
Code                 object
Year                 int64
Annual change in population    float64
Annual population change (Projected)  float64
dtype: object
```

```
In [7]: df.describe()
```

	Year	Annual change in population	Annual population change (Projected)
count	38400.000000	1.868800e+04	1.971200e+04
mean	2025.500000	2.063019e+06	8.375481e+05
std	43.300872	9.440859e+06	5.816745e+06
min	1951.000000	-3.315935e+06	-2.291035e+07
25%	1988.000000	1.781000e+03	-9.742750e+03
50%	2025.500000	4.689800e+04	9.350000e+01
75%	2063.000000	3.021128e+05	1.411160e+05
max	2100.000000	9.337137e+07	7.206736e+07

```
In [8]: df.isnull().sum()
```

```
Entity                0
Code                 1200
Year                0
Annual change in population    19712
Annual population change (Projected)  18688
dtype: int64
```

```
In [9]: df["Entity"].nunique()
```

```
256
```

- 1. ¿Cuántas filas y columnas tiene el dataset?** 38,400 filas y 5 columnas.
- 2. ¿Qué tipo de datos contiene la variable Year?** int64.
- 3. ¿Existen valores faltantes?** Sí, en las columnas Code (1,200), Annual change in population (19,712) y Annual population change Projected (18,688).
- 4. ¿Cuántas entidades diferentes contiene el dataset?** 256 entidades.

Actividad 3 — Estadística descriptiva

Filtro: **Afghanistan**, período **2005–2023**.

```
In [10]: df_A = df[(df["Year"] >= 2005) & (df["Year"] <= 2023) & (df["Entity"] == "Afghanist")
df_A.head(19)
```

	Entity	Code	Year	Annual change in population	Annual population change (Projected)
54	Afghanistan	AFG	2005	843919.0	NaN
55	Afghanistan	AFG	2006	1019525.0	NaN
56	Afghanistan	AFG	2007	485752.0	NaN
57	Afghanistan	AFG	2008	572779.0	NaN
58	Afghanistan	AFG	2009	983471.0	NaN
59	Afghanistan	AFG	2010	817986.0	NaN
60	Afghanistan	AFG	2011	1063620.0	NaN
61	Afghanistan	AFG	2012	1212328.0	NaN
62	Afghanistan	AFG	2013	1062673.0	NaN
63	Afghanistan	AFG	2014	1169819.0	NaN
64	Afghanistan	AFG	2015	1039239.0	NaN
65	Afghanistan	AFG	2016	868846.0	NaN
66	Afghanistan	AFG	2017	988328.0	NaN
67	Afghanistan	AFG	2018	1054099.0	NaN
68	Afghanistan	AFG	2019	1113085.0	NaN
69	Afghanistan	AFG	2020	1212852.0	NaN
70	Afghanistan	AFG	2021	931433.0	NaN
71	Afghanistan	AFG	2022	578437.0	NaN
72	Afghanistan	AFG	2023	875915.0	NaN

```
In [11]: df_A.mean(numeric_only=True)

Year                2014.000000
Annual change in population    941795.052632
dtype: float64
```

```
In [12]: df_A.median(numeric_only=True)

Year                2014.0
Annual change in population    988328.0
dtype: float64
```

```
In [13]: df_A.var(numeric_only=True)
```

```
Year                3.166667e+01
Annual change in population  4.446413e+10
dtype: float64
```

```
In [14]: df_A.std(numeric_only=True)
```

```
Year                5.627314
Annual change in population  210865.183261
dtype: float64
```

```
In [15]: df_A.min(numeric_only=True)
```

```
Year                2005.0
Annual change in population  485752.0
dtype: float64
```

```
In [16]: df_A.max(numeric_only=True)
```

```
Year                2023.0
Annual change in population  1212852.0
dtype: float64
```

1. **¿Cuál es el valor de la media?** 941,795.05 personas.
2. **¿Cuál es el valor de la mediana?** 988,328 personas.
3. **¿Cuál es el valor de la desviación estándar?** 210,865.18 personas.
4. **¿Qué entidad presenta el valor mínimo y cuál el máximo?** Mínimo: 485,752 (Afghanistan, 2007). Máximo: 1,212,852 (Afghanistan, 2020).

Actividad 4 — Comprobación del entorno

```
In [17]: import sys
print("Python:", sys.version)
print("Pandas:", pd.__version__)
print("NumPy: ", np.__version__)
```

```
Python: 3.11.15 (main, Mar 3 2026, 09:26:23) [GCC 13.3.0]
Pandas: 3.0.2
NumPy: 2.4.4
```